Q1. How embedded systems and sensors work together? Please explain.

Durgam Veena

A1. Embedded systems and sensors are perfect combinations that are working seamlessly in many applications. Sensing technology is evolving at a very fast pace with miniaturisation of microprocessors, microcontrollers, and other modules. Sensing technology plays an important role in many embedded systems.

There is a wide variety of sensors available for an embedded application to choose from. The most common/popular types of sensors interfaced with embedded systems include temperature, pressure, humidity, motion, infrared (IR), ultrasonic, Bluetooth, Internet of Things (IoT), among others.

Modern embedded systems are often based on microcontrollers. Sensors are either inbuilt or interfaced with I/O pins of microcontrollers. For example, microcontrollers like MSP430F6723 and MSP430F2013 have inbuilt temperature sensors. The sensor input is read by the microcontroller, processed, and the output data is displayed on output display devices such as LCDs, OLEDs, 7-segment LEDs connected to the microcontroller.

There are also different types of embedded sensors (embedded systems with sensors) available in the market. These sensors are actually part of embedded systems that interact with the surrounding environment.

With modern sensors and software components, embedded sensors are an active and growing area of embedded computing. For example, MICA2 Mote is a popular wireless network embedded sensor module. It supports 868/916MHz, 433/315MHz multi-channel transceivers with extended range. It can be interfaced with different sensors to measure light, temperature, RH, barometric pressure, acceleration/seismic, acoustic, magnetic, and other parameters.

Q2. What is OpenHab technology? How can I build automation on the openHAB platform?

Pamarthi Kanakaraja

A2. Open Home Automation Bus or openHAB is an open source home automation project started in 2010. It is a Java-based project supporting various home automation technologies and devices. It requires a Java virtual machine and can be deployed on servers running various operating systems. Its website claims that it can run on any operating system such as Linux, macOS, Windows, Raspberry Pi, PINE64, Docker, Synology, and others.

Fig. 1: openHAB home page

All the controlling actions for automation are triggered by rules, voice commands, or controls on the openHAB user interface. A binding in openHAB is an additional package used to connect or interact with other devices. It is like a plugin that provides support for different smart devices. For example, bindings include Bluetooth, ZigBee, Nest thermostat, etc.

To build an automation project on openHAB, first open their website *https://www.openhab.org/*. You need to click on Get Started or Demo option, as shown in Fig. 1. (If you are new to openHAB, you may first check the demo. There is also online documentation for beginners to get started.) Then you need to choose the operating system you would want to start with. It is strongly recommended to read the software installation guide given on their website. You may also get help from active users from the online community website at *https://community.openhab.org*

Answers compiled by Sani Theo, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: **editsec@efy.in**) and should include name and address of the sender.